

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number a is **190%** greater than the number b . The number b is **80%** less than **24**. What is the value of a ?

Answer

- A. **9.12**
- B. **13.92**
- C. **26.40**
- D. **36.48**

Correct Answer: B

Rationale

Choice B is correct. It's given that the number b is **80%** less than **24**. It follows that b is equal to **24** minus **80%** of **24**, which can be written as $b = 24 - (\frac{80}{100})24$. This is equivalent to $b = 24 - 0.8(24)$, or $b = 4.8$. It's also given that the number a is **190%** greater than the number b . It follows that a is equal to b plus **190%** of b . Since $b = 4.8$, this can be written as $a = 4.8 + (\frac{190}{100})4.8$. This is equivalent to $a = 4.8 + 1.9(4.8)$, or $a = 13.92$.

Choice A is incorrect. This would be the value of a if a were **190%** of b , not **190%** greater than b .

Choice C is incorrect. This is **(190 – 80)%** of **24**.

Choice D is incorrect. This would be the value of a if b were **80%** of **24**, not **80%** less than **24**, and a were **190%** of b , not **190%** greater than b .